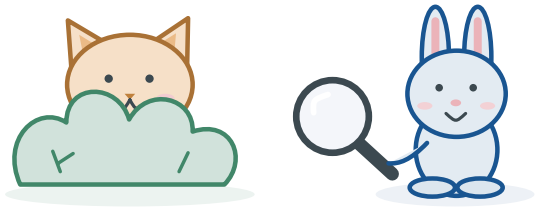
More applications of Nash Equilibrium



I. Hide-and-Seek: Choosing a Location

A hider and a seeker independently choose one of $n \geq 2$ locations. The seeker wins if their choices match; the hider wins otherwise. A win gives payoff 1 and a loss gives payoff 0.

For three locations, entries list the hider's payoff, then the seeker's:

Hider / Seeker	L	C	R
L	$0, 1$	$1, 0$	$1, 0$
C	$1, 0$	$0, 1$	$1, 0$
R	$1, 0$	$1, 0$	$0, 1$

No pure-strategy profile is an equilibrium. At a match the hider can move; without a match the seeker can change locations and win.

Making Every Location Equally Attractive

Uniform randomization makes every location equally attractive to both players. To see why, let p_j be the probability of hiding at location j, and let q_j be the probability of searching there. The expected payoffs from choosing location j are

$$u_H(j, q) = 1 - q_j, \qquad u_S(p, j) = p_j.$$

These payoffs make the hider prefer the least searched location and the seeker prefer the most likely hiding location.

To make every location a best response, suppose both players choose each with probability $1/n$. Under these mixtures, their expected payoffs are

$$u_H = 1 - \frac{1}{n}, \qquad u_S = \frac{1}{n}.$$

Thus, uniform randomization by both players is a Nash equilibrium.

Why Uniform Mixing Is Necessary

To see why every equilibrium requires uniform mixing, suppose $q_j = 0$ for some location. Hiding there guarantees payoff 1, so the hider would use only locations that are never searched.

- Against such hiding choices, the seeker would never catch the hider. Searching a location with positive hiding probability would do better, so an equilibrium must assign positive search probability everywhere.

Because the seeker uses every location, each must give the seeker the same payoff p_j. Since these probabilities sum to one, $p_j = 1/n$.

The hider therefore uses every location too. For these choices to be optimal, all q_j must also be equal. Thus, $q_j = 1/n$, establishing that uniform mixing by both players is the unique equilibrium.

Predictability Can Be Exploited

A hider can exploit unequal search probabilities by choosing a less searched location. For example, suppose the three locations are searched with probabilities

$$(q_L, q_C, q_R) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right).$$

This search pattern gives the hider payoffs $1/2, 3/4, 3/4$ from L, C, R. So hiding at C or R, or mixing between them, is a best response.

However, hiding only at C is not part of an equilibrium: the seeker would then prefer to search C with certainty.

This incentive to change locations distinguishes hide-and-seek from coordination. A conspicuous location may attract both players in a coordination game; here, the hider wants to avoid the seeker.

Example: Targets with Different Values

An attacker chooses one of four targets; a defender simultaneously chooses one target to protect. Unlike the identical hiding locations, these targets have different values: target v has value v. If the defender protects the attacked target, both get zero. Otherwise, the attacker gets its value v and the defender gets $-v$.

Rows are attacks, columns are defenses, and entries list the attacker's payoff followed by the defender's payoff.

	D_1	D_2	D_3	D_4
A_1	$0, 0$	$1, -1$	$1, -1$	$1, -1$
A_2	$2, -2$	$0, 0$	$2, -2$	$2, -2$
A_3	$3, -3$	$3, -3$	$0, 0$	$3, -3$
A_4	$4, -4$	$4, -4$	$4, -4$	$0, 0$

Which Target Can We Eliminate?

The attacker can do better than attacking target 1 by mixing equally over A_2, A_3, A_4. To see why, compare the payoffs against each defense:

Defense	D_1	D_2	D_3	D_4
Payoff from A_1	0	1	1	1
Payoff from the mixture	3	7/3	2	5/3

The mixture gives a strictly higher payoff in every column, so we can eliminate A_1 as strictly dominated.

Once A_1 is removed, defending target 1 leaves every remaining attack unprotected, giving payoff $-v$ against A_v. Mixing equally over D_2, D_3, D_4 instead gives $-2v/3 > -v$. So we can also eliminate D_1.

The Remaining Three Targets

Removing A_1 and D_1 leaves targets with values 2, 3, and 4. The payoffs and timing remain the same.

Rows are attack choices and columns are defense choices:

	D_2	D_3	D_4
A_2	$0, 0$	$2, -2$	$2, -2$
A_3	$3, -3$	$0, 0$	$3, -3$
A_4	$4, -4$	$4, -4$	$0, 0$

A pure-strategy profile cannot be an equilibrium: an undefended attack invites a change in defense; a defended attack invites a different target.

Making Three Attack Choices Indifferent

To make all three attacks optimal, the defender must equalize their expected payoffs. If target v is defended with probability q_v, an attack succeeds with probability $1 - q_v$, so we require

$$2(1 - q_2) = 3(1 - q_3) = 4(1 - q_4) = U.$$

Here U is the common attack payoff, so $q_v = 1 - U/v$. Summing these defense probabilities to one determines U:

$$3 - U\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) = 1, \qquad U = \frac{24}{13}.$$

Substituting this payoff into $q_v = 1 - U/v$ gives the defense probabilities

$$(q_2, q_3, q_4) = \left(\frac{1}{13}, \frac{5}{13}, \frac{7}{13}\right).$$

The most valuable target receives the most protection.

Making Three Defense Choices Indifferent

Next, let's make all three defenses optimal. If target v is attacked with probability p_v, protecting it avoids expected damage vp_v. Indifference therefore requires

$$2p_2 = 3p_3 = 4p_4, \qquad p_2 + p_3 + p_4 = 1.$$

So the attack probabilities are

$$(p_2, p_3, p_4) = \left(\frac{6}{13}, \frac{4}{13}, \frac{3}{13}\right).$$

Against these attack probabilities, each defense gives payoff $-24/13$. Likewise, each attack gives payoff $24/13$ against the defense mixture q. Since all available actions are best responses, these mixtures form an equilibrium in the reduced game.

- Notice that the most valuable target is attacked least often. Its higher value means that fewer attacks suffice to make protection worthwhile.

Checking the Original Four-Target Game

Finally, let's check whether the equilibrium of the reduced game is also an equilibrium of the original four-target game. The proposed mixtures are

$$p = \left(0, \frac{6}{13}, \frac{4}{13}, \frac{3}{13}\right), \qquad q = \left(0, \frac{1}{13}, \frac{5}{13}, \frac{7}{13}\right)$$

for attacks and defenses, respectively. The used attacks give payoff 24/13, and the used defenses give payoff $-24/13$. A deviation to target 1 must therefore be compared with these payoffs.

Attacking target 1 gives only $1 < 24/13$. Defending target 1 leaves all attacks unprotected and gives

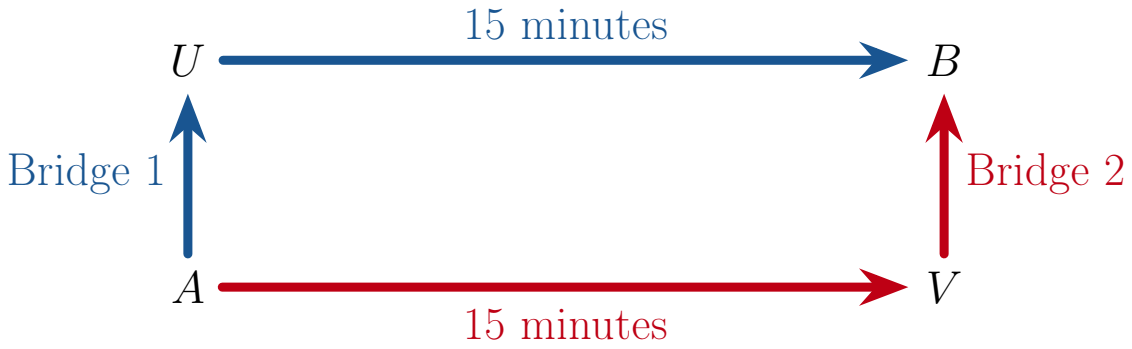
$$-2\frac{6}{13} - 3\frac{4}{13} - 4\frac{3}{13} = -\frac{36}{13} < -\frac{24}{13}.$$

Neither excluded action improves the relevant payoff. Thus, the mixtures are mutual best responses in the original four-target game.

II. The Transportation Paradox

There are 1,000 drivers traveling from A to B. Each simultaneously chooses a route and minimizes travel time. A bridge carrying F cars takes $F/100$ minutes; each ordinary road takes 15 minutes.

The two routes each combine one bridge with one ordinary road:



Route 1 is $A \to U \to B$; Route 2 is $A \to V \to B$.

Equilibrium Before the New Road

Equilibrium initially splits drivers equally between the routes. Let $R_1+R_2 = 1000$ count all drivers, including the one whose time we calculate:

$$T_1 = 15 + \frac{R_1}{100}, \qquad T_2 = 15 + \frac{R_2}{100}.$$

At $R_1 = R_2 = 500$, everyone takes 20 minutes. A driver switching routes would raise the other route's count to 501 and take 20.01 minutes, so no one gains.

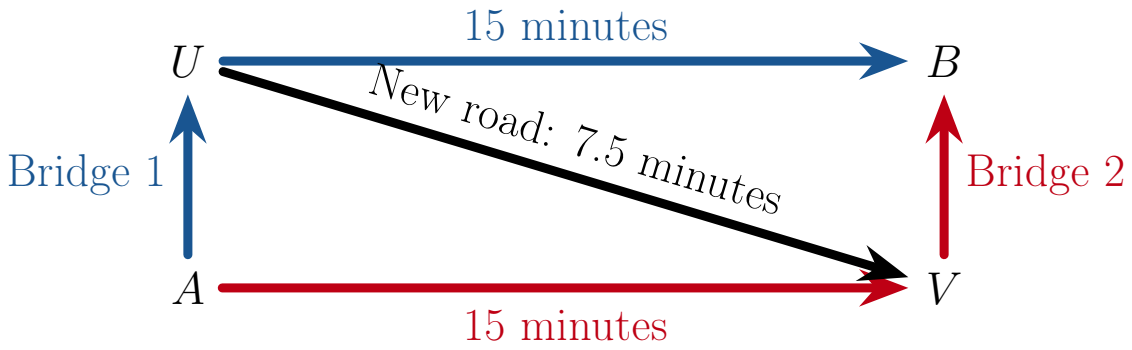
By contrast, any unequal split has at least 501 drivers on the busier route and at most 499 on the other. Switching to the quieter route therefore reduces a driver's travel time.

Thus, the unique pure-strategy equilibrium route counts are $(500, 500)$.

- Many strategy profiles produce these counts.

A Faster Road Creates a Third Route

Add a one-way road from U to V taking 7.5 minutes. Drivers can now choose Route 3, $A \to U \to V \to B$, which uses both bridges. The original roads remain available.



Starting from the old split, a driver can switch to Route 3 and take $5 + 7.5 + 5.01 = 17.51$ minutes instead of 20. This profitable deviation means that the old profile is no longer an equilibrium.

More Drivers on Both Bridges

To find an equilibrium with the new road, we must account for the traffic that Route 3 adds to both bridges. With $R_1 + R_2 + R_3 = 1000$, Bridge 1 carries $R_1 + R_3$ cars and Bridge 2 carries $R_2 + R_3$. Travel times are

$$\begin{aligned} T_1 &= 15 + (R_1 + R_3)/100, \\ T_2 &= 15 + (R_2 + R_3)/100, \\ T_3 &= 7.5 + (R_1 + R_3)/100 + (R_2 + R_3)/100. \end{aligned}$$

Seek a symmetric candidate with $R_1 = R_2 = r$ and $R_3 = 1000 - 2r$. Equal travel times require

$$15 + \frac{1000 - r}{100} = 7.5 + 2\frac{1000 - r}{100}.$$

This gives $(R_1, R_2, R_3) = (250, 250, 500)$. Both bridges carry 750 cars, and all three routes take 22.5 minutes.

Checking Each Driver's Incentive

Let's check for profitable deviations at $(250, 250, 500)$. The alternative travel times include each driver's effect on bridge traffic:

Current route	Alternative route	Time after moving
1	2 or 3	22.51
2	1 or 3	22.51
3	1 or 2	22.50

Since no deviation improves on 22.5 minutes, these route choices form a Nash equilibrium. Yet everyone takes longer than the original 20 minutes.

The extra delay arises because drivers use both bridges without accounting for the congestion they impose on others. Thus, adding a road can make everyone worse off in equilibrium: the **Braess paradox**.

A Faster Link Can Worsen Congestion

If the new link takes 5 minutes, all 1,000 drivers using Route 3 is an equilibrium. Each takes $10 + 5 + 10 = 25$ minutes. Switching to Route 1 or 2 still takes $10 + 15 = 25$ minutes, so no driver gains.

The equilibrium examples give the following comparison:

Network	Route counts (R_1, R_2, R_3)	Minutes
Original	$(500, 500, 0)$	20
7.5-minute link	$(250, 250, 500)$	22.5
5-minute link	$(0, 0, 1000)$	25

The original 20-minute allocation remains feasible: a planner could keep all drivers on the old routes. Thus, the best achievable outcome cannot worsen, although individual route choices can produce a worse equilibrium.

Example: Cheonggyecheon Restoration

Seoul removed an elevated road to restore Cheonggyecheon in 2003–2005. The photos show the road before removal and the restored stream.

Before: elevated road (1969)



National Archives of Korea, KOGL Type 1

After: restored stream



lumoplank, CC0 1.0

Travelers made fewer road trips and used the subway more. Thus, adjustment extended beyond route choices to transport modes.

Sources: Seoul Metropolitan Government; Chung, Hwang, and Bae (2012), *Transport Policy*.

III. Stripped-Down Poker: The Rules

A professor and a student each put one token into a pot. The professor privately draws K or J, each with probability 1/2. The student has an intermediate card: it loses to K and beats J.

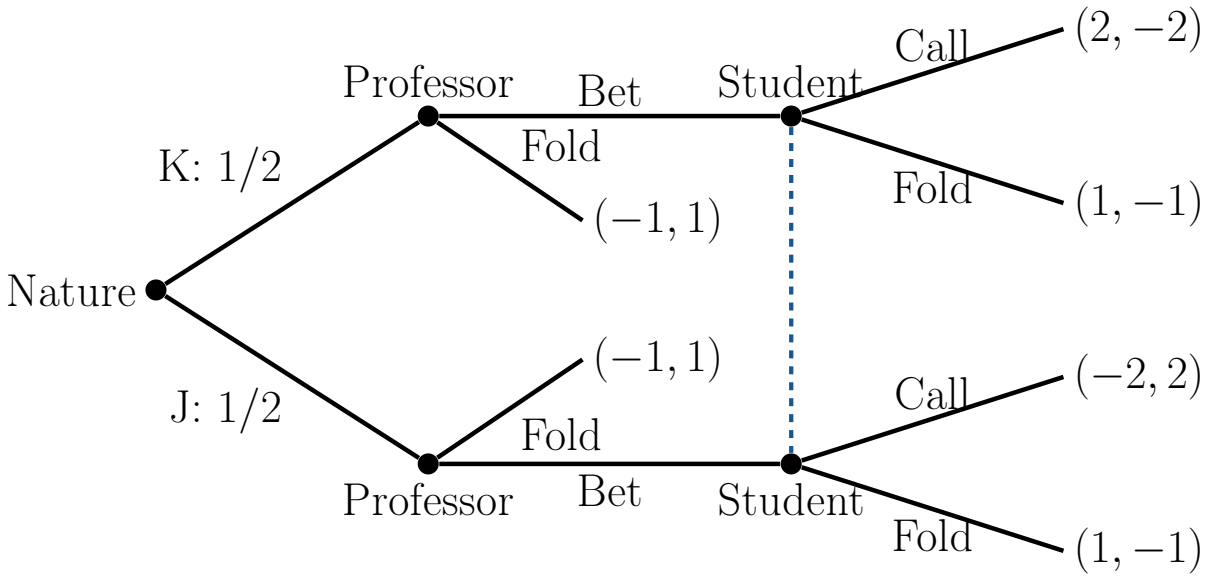
The professor can fold, losing the initial token, or bet one more token. After a bet, the student can fold or call by adding one token.

- If either player folds, the other wins the pot. Net payoffs are +1 to the winner and −1 to the player who folds.
- If the student calls, the higher card wins. Net payoffs are +2 to the winner and −2 to the loser.

When making this choice, the student observes the bet but not the professor's card.

The Game Tree and the Hidden Card

The tree begins with Nature choosing the professor's card. Because the student cannot observe that card, the two student nodes share one information set, joined by a dashed line. Terminal payoffs list the professor's payoff followed by the student's payoff.



Strategies for Each Possible Card

The professor has two information sets, one for each card. A pure strategy specifies an action at both, even though only one card is drawn. The first letter below is the action with K, and the second is with J.

Strategy	With K	With J
BB	Bet	Bet
BF	Bet	Fold
FB	Fold	Bet
FF	Fold	Fold

By contrast, the student has one information set and only two pure strategies: Call and Fold.

- The student cannot plan to "call against J but fold against K".

Averaging over the Two Cards

To represent the poker game in normal form, we average payoffs over Nature's equally likely K and J draws for each pair of strategies. The payoff of the professor appears before the payoff of the student.

Professor / Student	Call	Fold
BB	$0, 0$	$1, -1$
BF	$1/2, -1/2$	$0, 0$
FB	$-3/2, 3/2$	$0, 0$
FF	$-1, 1$	$-1, 1$

- For example, at (BF, Call), K leads to a called bet and J leads to a fold by the professor. The professor's expected payoff is

$$\frac{1}{2}(2) + \frac{1}{2}(-1) = \frac{1}{2}.$$

Strong Cards Are Always Bet

The professor must bet with K in equilibrium, because folding with K is strictly dominated. Specifically, BB strictly dominates FB: changing the action with K from Fold to Bet improves the payoff against both student strategies. Likewise, BF strictly dominates FF.

After removing FB and FF, the relevant payoff matrix is

	Call	Fold
BB	$0, 0$	$1, -1$
BF	$1/2, -1/2$	$0, 0$

Against Call, the professor prefers BF; against Fold, BB. Against BB, the student prefers Call; against BF, Fold. So there is no pure-strategy Nash equilibrium.

How Often Should the Student Call?

The student must call with a probability c that makes BB and BF equally attractive. The professor's expected payoffs are

$$u_P(\text{BB}, c) = 1 - c, \qquad u_P(\text{BF}, c) = \frac{c}{2}.$$

For the professor to mix between BB and BF, these payoffs must be equal:

$$1 - c = \frac{c}{2}, \qquad c = \frac{2}{3}.$$

– To interpret this calling probability, recall that betting always beats folding with K. With J, however, betting against $c = 2/3$ gives payoff

$$\frac{2}{3}(-2) + \frac{1}{3}(1) = -1,$$

exactly the payoff from folding. Thus, calling with probability 2/3 makes the professor indifferent about bluffing with J.

How Often Should the Professor Bluff?

In turn, the professor must bluff often enough to make the student indifferent between calling and folding. Let b be the probability of choosing BB, with probability $1-b$ on BF. The professor always bets with K and bets with J with probability b.

This bluffing probability determines the student's expected payoffs from Call and Fold:

$$u_S(b, \text{Call}) = -\frac{1-b}{2}, \qquad u_S(b, \text{Fold}) = -b.$$

For the student to mix between Call and Fold, these payoffs must be equal:

$$-\frac{1-b}{2} = -b, \qquad b = \frac{1}{3}.$$

Thus, the professor mixes $(1/3, 2/3)$ over (BB, BF), while the student mixes $(2/3, 1/3)$ over (Call, Fold). Expected payoffs are $(1/3, -1/3)$.

Why Bluffing Is Part of Equilibrium

At equilibrium, the professor bets with K with probability 1 and with J with probability 1/3. The probabilities of the two kinds of bet are

$$\Pr(\text{K and bet}) = \frac{1}{2}, \qquad \Pr(\text{J and bet}) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}.$$

So among bets, three quarters carry K and one quarter carry J. On observing a bet, calling gives the student

$$\frac{3}{4}(-2) + \frac{1}{4}(2) = -1,$$

the same payoff as folding. With fewer bluffs, calling would be worse than folding; with more bluffs, calling would be better.

Finally, returning to the excluded strategies cannot help the professor: FB and FF both give payoff -1 against $c = 2/3$, below the equilibrium payoff 1/3.

IV. Mixed Chicken: Who Yields?

Two players simultaneously choose an aggressive action H or yield with D. The payoff of player 1 is listed before that of player 2.

	H	D
H	$0, 0$	$3, 1$
D	$1, 3$	$2, 2$

Against H, each player prefers D; against D, each prefers H. Thus, (H, D) and (D, H) are the two pure-strategy equilibria.

Can the players also be mutual best responses while randomizing?

The Symmetric Mixed-Strategy Equilibrium

For both players to randomize, each must be indifferent between H and D. Let p and q be their respective probabilities of H. Player 1 earns

$$u_1(H, q) = 3(1 - q), \qquad u_1(D, q) = q + 2(1 - q) = 2 - q.$$

Indifference requires $3 - 3q = 2 - q$, so $q = 1/2$. Similarly, player 2 is indifferent when $p = 1/2$.

Both mixing equally is therefore a Nash equilibrium. Each receives expected payoff 3/2; any unilateral mixture gives the same payoff against the opponent's equilibrium mixture.

There cannot be an equilibrium in which only one player mixes: against a pure action, the other player has a unique best response. Hence, the two pure-strategy equilibria and this mixed-strategy equilibrium are the only equilibria.

Randomization Does Not Remove the Conflict

At the mixed-strategy equilibrium, the players randomize independently. The probabilities of the four outcomes are

Outcome	(H, H)	(H, D)	(D, H)	(D, D)
Probability	1/4	1/4	1/4	1/4
Total payoff	0	4	4	4

Mutual aggression occurs with probability 1/4, reducing the expected total payoff to 3. This is below the total of 4 at either pure-strategy equilibrium, where one player yields and the other is aggressive.

The mixed-strategy equilibrium treats players alike but risks mutual aggression. The pure-strategy equilibria avoid this risk by assigning different roles. Equilibrium alone does not select among them.

V. Cournot Competition with Many Firms

There are $n \geq 1$ identical firms producing the same good. Each firm i simultaneously chooses a quantity $q_i \geq 0$. If total output is $Q = \sum_{i=1}^{n} q_i$, the market price is

$$P = a - bQ, \qquad a > c > 0, \quad b > 0.$$

Each firm has constant marginal cost c and no fixed cost. Since every unit produced is sold at the market price, firm i earns profit

$$\pi_i(q_i, q_{-i}) = (a - bQ - c)q_i.$$

The number of firms is fixed within each game. Comparing equilibria across different values of n shows how the number of competitors changes output, price, and profit.

Responding to the Other Firms' Total Output

Firm i cares about its rivals' combined output, $Q_{-i} = \sum_{j \neq i} q_j$. Holding that output fixed, its profit is

$$\pi_i = (a - c - bQ_{-i})q_i - bq_i^2.$$

The marginal profit from increasing q_i is $a - c - bQ_{-i} - 2bq_i$. Since the second derivative is $-2b < 0$, the best response, including the nonnegativity constraint, is

$$\text{BR}_i(Q_{-i}) = \max\left\{0, \frac{a - c - bQ_{-i}}{2b}\right\}.$$

- Thus, more output from rivals reduces the firm's optimal output. Quantities are strategic substitutes, just as in the two-firm model.

Why All Firms Produce the Same Quantity

Every firm produces the same positive quantity in equilibrium. To establish this, first note that at least one firm must produce: if all choose zero, producing a small positive quantity is profitable. For any firm with $q_i > 0$, its best-response condition can be written as

$$a - c - bQ = bq_i > 0.$$

If another firm produced zero, its marginal profit from starting production would be $a - c - bQ > 0$. So every firm must produce in equilibrium.

The displayed condition therefore applies to every firm. Its left-hand side is the same for all firms, which implies $q_i = q_j$ for every pair. Writing their common output as q^* gives

$$a - c - bnq^* = bq^*, \qquad q^* = \frac{a - c}{b(n+1)}.$$

More Firms Bring Price Closer to Cost

The common equilibrium quantity determines how the number of firms affects market outcomes. Substituting $q^* = (a-c)/[b(n+1)]$ gives

$$Q^* = \frac{n(a-c)}{b(n+1)}, \qquad P^* = c + \frac{a-c}{n+1}, \qquad \pi_i^* = \frac{(a-c)^2}{b(n+1)^2}.$$

- These quantities satisfy every firm's best response. Together with the preceding argument, this establishes the unique pure-strategy equilibrium.

As n increases, each firm produces less, but total output rises. That increase in total output lowers the market price and every firm's profit. In the limit,

$$Q^* \longrightarrow \frac{a-c}{b}, \qquad P^* \longrightarrow c, \qquad \pi_i^* \longrightarrow 0.$$

Thus, competition among many quantity-setting firms approaches the competitive price even though each firm maximizes its own profit.

From One Firm to Many Competitors

Keep $a = 1000$, $b = 1$, and $c = 100$, so demand and costs match the numerical duopoly. Changing only the number of firms gives:

Firms n	Each q_i^*	Total Q^*	Price P^*	Each π_i^*
1	450	450	550	202,500
2	300	600	400	90,000
4	180	720	280	32,400
9	90	810	190	8,100

With one firm, the equilibrium is the monopoly choice. With more firms, each chooses output to maximize its own profit and ignores the resulting losses to competitors. Their combined output exceeds the monopoly output.

VI. Differentiated-Product Pricing

Two firms sell differentiated products and simultaneously choose prices $p_1, p_2 \geq 0$. Each has zero production cost and supplies its demand:

$$D_1 = \max\{0, 10 - p_1 + p_2\}, \qquad D_2 = \max\{0, 10 - p_2 + p_1\}.$$

Firm i earns revenue, and hence profit, $\pi_i = p_i D_i$.

- The maximum with zero means that a sufficiently high price gives no sales.

Product differentiation lets a firm retain customers when its rival charges a slightly lower price. Where demand is positive, a one-unit rise in a firm's own price reduces its sales by one unit; a one-unit rise in its rival's price increases its sales by one unit.

Finding a Best Response in Prices

To find a firm's best response, we maximize its profit holding the rival's price $p_j \geq 0$ fixed. For $0 \leq p_i \leq 10 + p_j$, profit is

$$\pi_i = p_i(10 + p_j - p_i).$$

Its derivative is $10 + p_j - 2p_i$, and its second derivative is -2. The unique maximum in this range is therefore

$$p_i = \frac{10 + p_j}{2} = 5 + \frac{1}{2}p_j.$$

This price gives positive demand and profit. Prices above $10 + p_j$ give no sales and zero profit, so none can improve on it. Thus, the best-response function over all feasible prices is

$$\mathrm{BR}_i(p_j) = 5 + \frac{1}{2}p_j.$$

Solving for the Two Prices

At a Nash equilibrium, both prices must be best responses:

$$p_1 = 5 + \frac{1}{2}p_2, \qquad p_2 = 5 + \frac{1}{2}p_1.$$

Subtracting gives $p_1 - p_2 = -\frac{1}{2}(p_1 - p_2)$, so $p_1 = p_2$. Substituting this equality into either equation gives

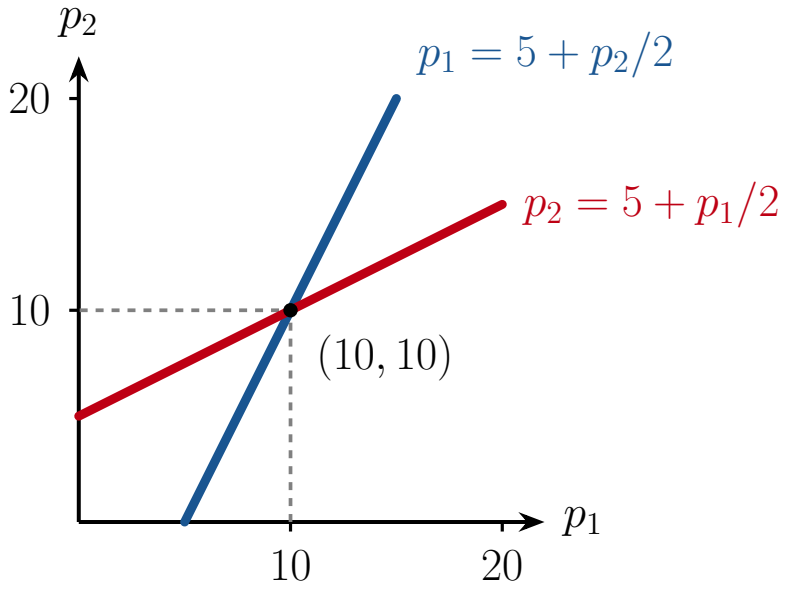
$$p_1^* = p_2^* = 10, \qquad D_1^* = D_2^* = 10, \qquad \pi_1^* = \pi_2^* = 100.$$

Since each best response is unique and the equations have only this solution, $(10, 10)$ is the unique pure-strategy Nash equilibrium.

Each firm prefers to charge 10 when it expects the other to charge 10. These individual incentives sustain positive equilibrium prices without an agreement between the firms.

Prices Are Strategic Complements

The best-response functions show that prices are strategic complements: each firm raises its optimal price when its rival charges more. Their intersection identifies the equilibrium in the graph below.



A higher rival price raises the firm's demand, explaining this upward response. By contrast, Cournot quantities are strategic substitutes: more output from rivals reduces the firm's optimal output.

Why Undercutting Does Not Pay

Undercutting is unprofitable here because the extra sales do not offset the lower price. If the rival charges 10, charging 10 sells 10 units and earns 100. Cutting its price to 9 raises sales to 11, but profit falls to $9(11) = 99$.

More generally, with the rival's price fixed at 10, profit for prices between 0 and 20 is

$$\pi_i = p_i(20 - p_i) = 100 - (p_i - 10)^2.$$

Any departure from 10 lowers profit in this range; prices above 20 give zero sales. Thus, gaining some customers by undercutting is not enough to compensate for the lower price on every unit sold.

In homogeneous-product Bertrand competition, undercutting captures all demand when the rival's price exceeds marginal cost. Here, product differentiation makes that gain gradual, allowing positive equilibrium markups even though firms compete in prices.

Takeaways

Equilibrium connects individual incentives to collective outcomes.

- In hide-and-seek, randomization prevents exploitation.
- In transportation, adding a road can increase congestion and make every driver worse off.
- In poker, equilibrium mixing makes the opponent indifferent.
- In Chicken, pure-strategy and mixed-strategy equilibria coexist and give different outcomes.
- In Cournot, more firms raise total output and bring the price closer to marginal cost.
- Differentiated products allow positive markups.
  Prices are strategic complements.